Random Forest Machine Learning Project

Diabetes Prediction Using the Pima Indians Diabetes Dataset

Classification | Random Forest | Kaggle real-world dataset | 768 records

Dataset & Problem Statement

Dataset: Pima Indians Diabetes Database

The task is to classify whether a patient has diabetes (Outcome = 1) or not (Outcome = 0) using eight numeric diagnostic/patient attributes.

Shape: 768 rows x 9 columns (8 predictors + 1 target)

Target classes: 500 no-diabetes, 268 diabetes

Data Preprocessing

- Loaded and explored the full dataset; checked data types, summary statistics and class balance.
- Converted invalid zero measurements in five medical fields to missing values.
- Applied median imputation inside an sklearn Pipeline to prevent data leakage.
- No categorical encoding was needed: all predictors are numeric and Outcome is already binary (0/1).
- Used a stratified 80/20 train-test split (614 training, 154 testing rows; random_state=42).

Model Used & Tuning

RandomForestClassifier in a Pipeline with median imputation. Hyperparameters were tuned using GridSearchCV with stratified 5-fold cross-validation, optimizing ROC-AUC. Best settings: n_estimators=300, max_depth=8, min_samples_leaf=2, max_features="sqrt", min_samples_split=2, random_state=42.

Evaluation Metrics

75.32%	69.05%	53.70%	60.42%	81.04%
Accuracy	Precision	Recall	F1-Score	ROC-AUC

Confusion matrix (test set): TN=87, FP=13, FN=25, TP=29.

Best 5-fold CV ROC-AUC: 0.8304 | Mean 5-fold CV accuracy: 0.7642.

Key Findings

- Glucose was the most important feature (31.77%), followed by BMI (16.57%) and Age (12.37%).
- The tuned model achieved an ROC-AUC of 0.8104 on the held-out test set, showing useful discrimination between the two classes.
- Recall for the diabetes class was 53.70%, so the model still misses some positive cases and should not be treated as a clinical diagnostic tool.

Conclusion

A complete Random Forest classification workflow was implemented successfully: exploration, cleaning, median imputation, stratified splitting, model training, hyperparameter tuning, prediction, multi-metric evaluation, cross-validation, feature importance visualization and Joblib model saving. The results show that glucose-related information is the dominant predictor in this dataset, while the held-out metrics indicate room for improved sensitivity.

Kaggle source:
https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database